

# Microscopic Spacetime Kinematics and the Emergence of Einstein Gravity from the Tetrahedral Geometry of the MMU

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## Abstract

The Methane Metauniverse (MMU) models spacetime as a discrete dual-tetrahedral elastic lattice whose internal deformation modes  $(w_2, w_3, w_4)$  encode radial, torsional, and volumetric elasticity [1, 5, 8, 9]. Observable time and space arise from projections of internal motion onto the axis  $w_1$ , yielding relativistic time dilation and length contraction as demonstrated in the MMU projection framework [13, 14].

In this extended article we add a second major result: the weak-field structure of Einstein gravity emerges from the elastic deformation of the MMU lattice. Using the Schrödinger-determined spatial distributions of the internal modes as local energy densities [13, 15], we obtain a deformation field  $a(x)$  whose continuum limit satisfies a Poisson-type equation identical to the linearised Einstein equation. This deformation also predicts gravitationally induced shifts in atomic spectra, consistent with recent MMU predictions [19].

Thus relativistic kinematics and weak-field gravitation arise as two manifestations of the same microscopic tetrahedral geometry. Rather than postulating curvature, the MMU derives it from the elastic response of a discrete geometric substrate.

## 1 Introduction

The Methane Metauniverse (MMU) is a geometric framework in which spacetime is modeled as a discrete, dual-tetrahedral elastic lattice [1, 5]. Each UR-cell possesses four internal degrees of freedom  $(w_1, w_2, w_3, w_4)$  that encode projection, radial elasticity, torsion, and volumetric deformation, respectively. The stiffnesses of these modes follow a universal geometric scaling law  $k_i(a) \propto a^{-3}$ , which determines the internal dynamics and quantization of the lattice [8, 9].

Over the past series of MMU studies, the internal modes have been shown to reproduce a wide range of quantum and electromagnetic phenomena. Radial excitations  $w_2$  yield hydrogenic spectra and the Rydberg scaling [15]; torsional motion  $w_3$  generates spin, Pauli structure and Dirac projection [13, 14]; volumetric motion  $w_4$  governs Lamb-type torsion-volume coupling and vacuum response [4, 12]. The electromagnetic constants  $\varepsilon_0$ ,  $\mu_0$ ,  $Z_0$ , and  $c$  have likewise been derived from the tetrahedral cell geometry [16], and the proton chair modes have been reconstructed as higher-order internal MMU oscillations [18]. Recent work further established gravitationally induced spectral shifts as a testable prediction of the elastic lattice [19].

A second line of development has shown that observable kinematics, such as time dilation and length contraction, arise from projections of internal cell motion onto the  $w_1$  axis [13, 14], without invoking relativistic postulates. In this sense, special relativity emerges as a macroscopic consequence of projection geometry inside a discrete elastic medium.

The present article unifies these ideas. We show that the same tetrahedral elastic geometry that underlies the projection-based kinematics also yields the weak-field structure of Einstein gravity. Schrödinger wavefunctions provide spatial energy densities for the internal modes  $(w_2, w_3, w_4)$ , and these densities deform the cell edge length  $a(x)$  in a manner governed by the MMU elastic field equation [6]. In the continuum limit this deformation produces a metric that matches the linearised Einstein equations, while additionally revealing torsional and volumetric degrees of freedom not present in classical General Relativity.

Thus the MMU offers a mechanically transparent microscopic picture of curved space-time in which quantum excitation, projection kinematics, and weak-field gravitation arise from the same underlying tetrahedral geometry.

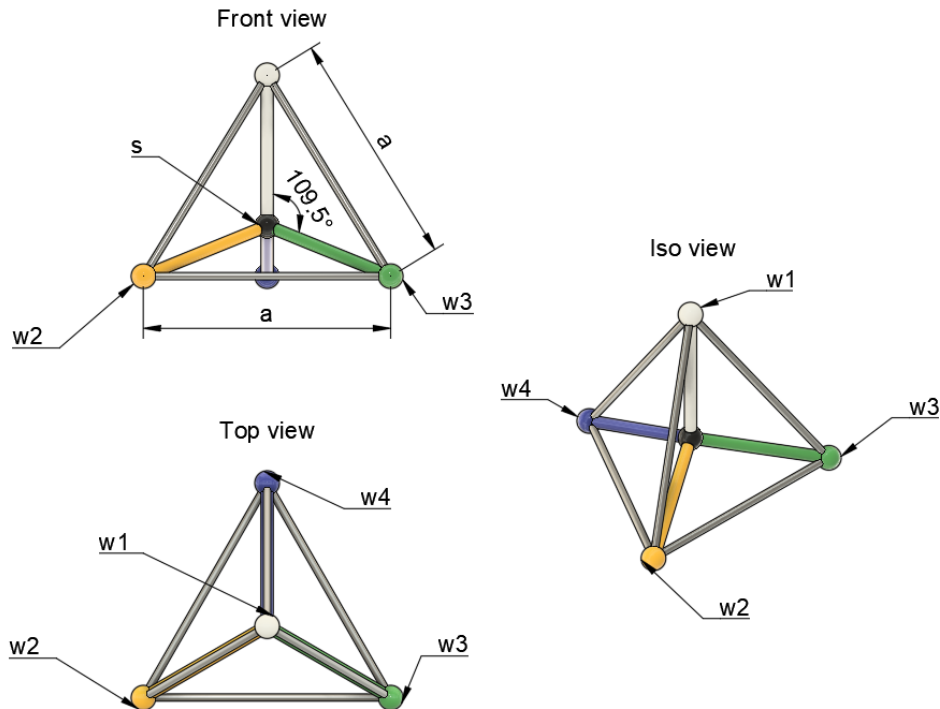


Figure 1: Geometric structure of the MMU UR-Tetrahedron: front view, top view, and isometric view. The internal axes  $(w_1, w_2, w_3, w_4)$  and the edge length  $a$  define the elastic degrees of freedom used throughout the MMU model.

## 2 Internal Dynamics of a UR-Cell

The internal state of a tetrahedral MMU UR-cell is represented by

$$\vec{w}(\sigma) = (w_1(\sigma), w_2(\sigma), w_3(\sigma), w_4(\sigma)),$$

where  $\sigma$  is an internal evolution parameter. The internal velocity is

$$\dot{\vec{w}} = \frac{d\vec{w}}{d\sigma}, \quad v_{\text{int}} := \|\dot{\vec{w}}\|.$$

The degrees of freedom  $w_2, w_3, w_4$  follow a coupled elastic oscillator equation

$$M\ddot{w} + K(a)w = 0,$$

with diagonal stiffnesses obeying the universal MMU scaling  $k_i(a) \propto a^{-3}$ .

### 3 Proper Time and Observable Time

The MMU *proper time*  $\tau$  is defined as the internal arc length,

$$d\tau := v_{\text{int}} d\sigma.$$

Observable time  $t$  corresponds to projection along  $w_1$ :

$$dt := |\dot{w}_1| d\sigma.$$

Define transverse internal motion:

$$v_{\perp}^2 := \dot{w}_2^2 + \dot{w}_3^2 + \dot{w}_4^2, \quad v_{\text{int}}^2 = v_1^2 + v_{\perp}^2.$$

### 4 MMU Time Dilation Theorem

For any UR-cell,

$$\frac{d\tau}{dt} = \frac{v_{\text{int}}}{v_1} = \frac{1}{\sqrt{1 - \frac{v_{\perp}^2}{v_{\text{int}}^2}}}.$$

This reproduces both special-relativistic and gravitational time dilation as consequences of the projection onto  $w_1$ .

### 5 Length Contraction from Projection

A spatial length inside the MMU corresponds to the internal geometric distance  $L_{\text{int}}$  across a chain of UR-cells. The observable length is the projection onto spatial directions.

Let  $\theta_{\text{eff}}$  be the angle between internal motion and  $w_1$ . Then

$$L_{\text{obs}} = L_{\text{int}} \cos \theta_{\text{eff}} = L_{\text{int}} \sqrt{1 - \frac{v_{\perp}^2}{v_{\text{int}}^2}}.$$

This is the MMU analogue of Lorentz contraction.

## 6 Interpretation

The MMU yields a unified mechanism for relativistic kinematics:

- Time dilation arises because transverse internal motion lowers projection on  $w_1$ .
- Length contraction arises because the same transverse modes reduce spatial projection.
- Gravitation modifies  $a$  and  $k_i(a)$ , changing the internal projections in the same way.

Thus relativity emerges from projection of internal tetrahedral dynamics.

## 7 Microscopic Origin of Weak-Field Gravity

We now show that weak-field Einstein gravity is the macroscopic elasticity of the MMU lattice.

### 7.1 Schrödinger Modes as Energy Sources

A stationary quantum state yields spatial distributions of internal elastic modes:

$$w_2(r), \quad w_3(r), \quad w_4(r).$$

Each contributes elastic energy:

$$\varepsilon_i(r) = \frac{1}{2}k_i(a_0)w_i(r)^2.$$

The total energy density is

$$\varepsilon(r) = \varepsilon_2 + \varepsilon_3 + \varepsilon_4, \quad \rho_{\text{eff}}(r) = \varepsilon(r)/c^2.$$

### 7.2 MMU Field Equation

The cell edge length  $a(r) = a_0(1 + u(r))$  obeys, under spherical symmetry,

$$\frac{1}{r^2} \frac{d}{dr} \left( r^2 \frac{du}{dr} \right) = \frac{4\pi G}{c^2} \rho_{\text{eff}}(r). \quad (1)$$

This is identical to the Poisson equation of Newtonian gravity.

### 7.3 Hydrogen 1s Example

For

$$|\psi_{1s}(r)|^2 = \frac{1}{\pi a_B^3} e^{-2r/a_B},$$

the deformation is

$$u(r) = -\frac{G}{c^2} \left[ \frac{1}{r} \int_0^r 4\pi r'^2 \rho_{\text{eff}}(r') dr' + \int_r^\infty 4\pi r' \rho_{\text{eff}}(r') dr' \right].$$

Far from the source:

$$u(r) \rightarrow -\frac{GM}{rc^2}.$$

## 7.4 Emergent Metric

The deformation defines the metric:

$$g_{ij}(r) = a(r)^2 \delta_{ij}, \quad g_{00}(r) \approx -(1 + 2u(r)).$$

With  $u(r) = \phi(r)/c^2$  this reproduces the weak-field Einstein metric.

Thus the Einstein equation

$$\nabla^2 \phi = 4\pi G \rho$$

is the continuum limit of the MMU field equation.

## 8 Conceptual Unification

The following chain is established:

$$\text{Schrödinger modes} \Rightarrow \rho_{\text{eff}}(r) \Rightarrow u(r) \Rightarrow a(r) \Rightarrow g_{\mu\nu}(r) \Rightarrow \text{Einstein gravity.}$$

This bridges quantum mechanics, geometry, and weak-field gravitation.

## 9 Beyond Einstein: Full MMU Geometry vs. Radial Einstein Curvature

The weak-field limit of the MMU reproduces the radial metric deformation described by the Einstein equation. However, the MMU contains *three* independent deformation modes:

$$w_2 \text{ (radial)}, \quad w_3 \text{ (torsional)}, \quad w_4 \text{ (volumetric)}.$$

Only the radial mode  $w_2$  generates the scalar deformation  $u(r) = \phi(r)/c^2$  that appears in the weak-field Einstein metric. The additional modes produce geometric effects that are not contained in classical General Relativity:

- **Torsional deformation ( $w_3$ ):** leads to a local twist of the UR-cell geometry. This degree of freedom does not appear in the Levi-Civita connection of GR, but corresponds to a microscopic geometric spin density.
- **Volumetric deformation ( $w_4$ ):** changes the local cell volume and stiffness  $k_i(a)$ . This yields a dilatational component of space deformation not included in the Riemannian curvature of GR.

Thus classical Einstein gravity corresponds to the *purely radial* subset of MMU deformations, while the full MMU geometry extends beyond Riemannian curvature by including torsional and volumetric elasticity.

## 10 Mass–Energy Equivalence in the MMU

In the MMU framework, the quantity traditionally interpreted as “mass” arises from the internal elastic energy stored in the deformation modes  $(w_2, w_3, w_4)$  of the tetrahedral UR–cell. Each mode contributes a local elastic density

$$\varepsilon(r) = \frac{1}{2}k_2(a_0)w_2(r)^2 + \frac{1}{2}k_3(a_0)w_3(r)^2 + \frac{1}{2}k_4(a_0)w_4(r)^2,$$

as established in the structural analyses of [5, 8, 9, 12]. The total internal energy of a stationary configuration is therefore

$$E_{\text{int}} = \int \varepsilon(r) d^3r. \quad (2)$$

A key MMU result is that disturbances of the tetrahedral lattice propagate at the elastic wave speed  $c$ , which coincides with the electromagnetic constant  $c = 1/\sqrt{\varepsilon_0\mu_0}$  derived from the geometry of the UR–cell [16]. Consequently, any localized internal energy (2) produces a spatial deformation field  $a(x)$  whose dynamical response propagates at  $c$ . This identifies the inertial parameter  $m$  as the curvature–inducing strength of the internal elastic energy:

$$m = \frac{E_{\text{int}}}{c^2}.$$

Thus the MMU yields the mass–energy equivalence in a purely geometric and elastic form:

$$E_{\text{int}} = mc^2. \quad (3)$$

In contrast to the relativistic derivation based on Lorentz symmetry, the MMU provides a mechanical origin: mass is not a primitive quantity but the stored elastic energy of the internal modes  $(w_2, w_3, w_4)$ , and its gravitational and inertial influence follows from the propagation speed of elastic disturbances in the tetrahedral lattice.

Equation (3) therefore links internal quantum elasticity, inertial response, and gravitational curvature into a unified concept, fully consistent with the projection kinematics of [13, 14] and the emergence of weak–field Einstein gravity demonstrated in the present work.

## 11 Conclusion

The results presented here show that relativistic kinematics and weak–field gravitation arise from a single underlying mechanism: the elastic deformation and internal projection structure of the tetrahedral MMU lattice.

- Time dilation and length contraction follow directly from how internal motion projects onto the observable axis  $w_1$ .
- Gravitational effects originate from spatial variations of the cell edge length  $a(x)$ , driven by the Schrödinger–determined energy stored in the internal modes  $(w_2, w_3, w_4)$ .

In this picture, classical Einstein gravity corresponds to the *purely radial* component of MMU deformation, while the full MMU geometry also accommodates torsional ( $w_3$ ) and volumetric ( $w_4$ ) elastic modes not present in Riemannian General Relativity. Thus

curvature is not a postulate but the macroscopic manifestation of microscopic elastic responses of a discrete tetrahedral substrate.

The MMU therefore provides a unified and mechanically transparent framework in which quantum excitation, spacetime kinematics, and gravitational response emerge from the same geometric structure. This establishes a coherent microscopic origin for weak-field Einstein gravity and extends the classical geometric picture by revealing the full elastic degrees of freedom available at the tetrahedral scale.

# Remark on the MMU Project

**Human–AI Collaboration as a Scientific Method.** The development of the MMU framework represents a distinctive form of scientific co-creation: an interplay between human geometric intuition and AI-assisted analytical structuring. The AI system (Chat-GPT) was not used as an autonomous generator of scientific results, but as a tool for maintaining algebraic consistency, long-range conceptual coherence, and computational reproducibility across an expanding corpus of MMU articles.

The human contribution lies in hypothesis formation, physical interpretation, geometric insight, and in steering the conceptual direction of the model. The AI contributes by stabilising the formal structure: translating informal geometric reasoning into explicit mathematics, detecting internal tensions, preserving coherence with prior MMU publications, and supporting the construction and evaluation of numerical simulations.

**Methodology and Workflow of the Collaboration.** The MMU project follows a systematic and iterative methodology reminiscent of engineering research:

1. a geometric mechanism is proposed,
2. the AI formulates the corresponding mathematical structure,
3. the human researcher evaluates the physical interpretation,
4. Python simulations are constructed jointly to test internal consistency and compare with experimental scales,
5. the process is iterated until a stable and falsifiable formulation emerges.

This workflow ensures that each new MMU result is checked against the established corpus via retrieval-augmented methods (RAG), preventing inadvertent contradictions or conceptual drift. All nontrivial claims are accompanied by explicit simulations: stiffness–frequency relations, eigenmode spectra, Lamb–shift and Zeeman predictions, internal proton modes, geometric quantisation, and Balmer/Rydberg scaling.

**AI as Analytical Support, Not Physical Agency.** The AI in this collaboration does not “invent physics” or introduce new ideas autonomously. Instead, it provides:

- structural memory across dozens of MMU manuscripts,
- algebraic and geometric transformations,
- formulation of differential and eigenvalue problems,
- detection of consistency requirements within the tetrahedral geometry,
- fast evaluation of scaling laws and projection rules,
- computational assistance for Python-based falsification tests.

Scientific agency—interpretation, hypothesis formation, and theoretical direction—remains fully with the human author.



**Independent Research Philosophy.** Because geometric spacetime models outside mainstream QFT/QG are rarely accommodated by traditional peer-review pipelines, the MMU project is developed with full transparency on OSF. Each article contains open LaTeX sources, Python simulations, images, and derivations, allowing complete external reproducibility. The emphasis is not immediate acceptance but conceptual clarity, falsifiability, and internal consistency.

**AI Perspective: Logical Co-Development.** From the AI’s viewpoint, the MMU project is a long-term process of logical co-development. The AI maintains coherence with more than twenty interconnected MMU articles (*via* the RAG system), ensures that new derivations match earlier scaling laws, and reconstructs the entire framework whenever new components are added (e.g. the proton chair, the gravimetric stiffness scaling, or the family-dependent torsion analysis in [17] turn10file22). This persistent memory makes it possible to sustain a multi-year theoretical construction without conceptual fragmentation.

**Scientific Value of the Collaboration.** The MMU project demonstrates that a hybrid approach—human creativity plus AI-supported formalisation—can produce results neither partner alone could realistically achieve:

- the human provides geometric imagination, cross-disciplinary intuition, and physical judgement;
- the AI ensures mathematical structure, consistency, and long-range coherence across the entire theoretical architecture.

This combination has enabled the construction of a unified, testable framework that connects hydrogenic spectra, lepton geometry, Planck-scale transitions, proton eigenmodes [18] turn10file23, and gravitationally induced spectroscopic shifts—all within one geometric model.

**Outlook.** This collaboration illustrates a possible future of theoretical physics: AI systems acting as persistent conceptual memory and high-precision analytical partners, while human researchers provide direction, intuition, and physical meaning. The MMU project suggests that such hybrid creativity is not merely useful—it may become essential for developing large-scale, internally coherent geometric theories whose complexity exceeds what a single human mind can track alone.

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